2026 May II
Module 1

Question N. 1

An artist made 12 pieces of pottery using 42 pounds of clay. The pottery included only bowls and vases. Each bowl used 2 pounds of clay and each vase used 4 pounds of clay. How many vases did the artist make?

A) 3

B) 5

C) 7

D) 9

Question N. 2

One gallon of paint will cover 10 square feet of a surface. A room has a total wall area of w square feet. Which equation represents the total amount of paint P, in gallons, needed to paint the walls of the room twice?

A) $P = 20w$

B) $P = 10w$

C) $P = \frac{w}{5}$

D) $P = \frac{w}{10}$

Question N. 3

x	y
$-\frac{4}{3}$	0
0	-128
4	0

The table shows three values of x and their corresponding values of y. There is a quadratic relationship between x and y. An equation that represents this relationship can be written as $y = 24x^2 - bx - 128$, where b is a constant. What is the value of b?

Question N. 4

The equation $E(t) = 5(1.2)^t$ gives the estimated number of employees at a certain business, where t is the number of years since the business opened. Which of the following is the best interpretation of the number 5 in this context?

A) The increase in the estimated number of employees each year

B) The estimated number of employees when the business opened

C) The percent increase in the estimated number of employees each year

D) The number of years the business has been open

Question N. 5

Line t in the xy-plane has a slope of $\frac{1}{7}$ and passes through the point $(35, -8)$. Which equation defines line t?

A) $y = -13x + \frac{1}{7}$

B) $y = \frac{x}{7} - 13$

C) $y = \frac{x}{7} - 8$

D) $y = 35x - 8$

Question N. 6

The area of a rectangular region is increasing at a rate of 210 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use 1 meter = 3.28 feet.)

A) 0.33

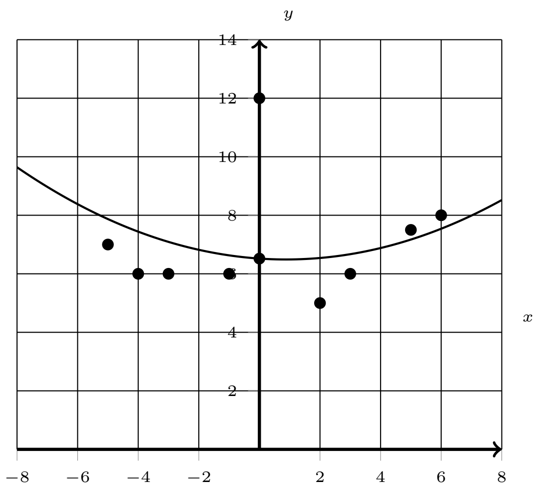
B) 1.07

C) 11.48

D) 19.52

Question N. 7

The scatterplot shows the relationship between x and y for the 9 data points in data set A.



A quadratic model for the data in data set A can be written as $y = 0.04x^2 - 0.07x + 6.59$. The graph of this model is shown in the xy-plane. The data point at $x = 0$ was found to be the result of a recording error and was removed from data set A to create data set B, consisting of the remaining 8 data points. A quadratic model for the data in data set B can be written as $y = ax^2 + bx + c$, where a, b, and c are constants. If the quadratic models for data sets A and B are calculated in the same way, which of the following statements must be true?
I. $a > 0.04$
II. $c < 6.59$

A) I only

B) II only

C) I and II

D) Neither I nor II

Question N. 8

$$f(x) = (2x - 7)(2x + 5)$$

What is the minimum value of the given function?

A) -36

B) $-\frac{5}{2}$

C) $\frac{1}{2}$

D) 2

Question N. 9

$$2x^2 + 18x + 2y^2 + 12y = \frac{15}{2}$$

The given equation defines a circle in the xy-plane. What is the radius of the circle?

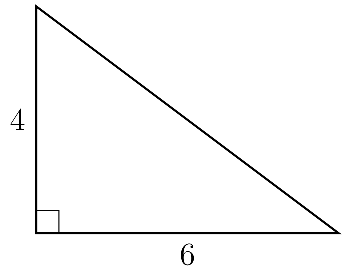
A) $\frac{\sqrt{30}}{2}$

B) $\sqrt{15}$

C) $\sqrt{33}$

D) $\frac{\sqrt{498}}{2}$

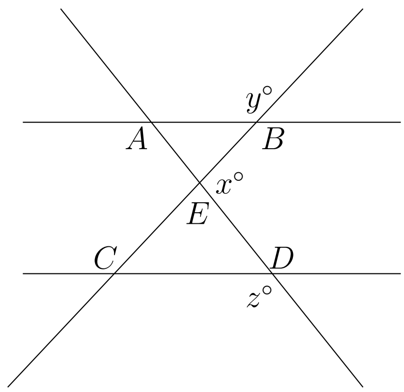
Question N. 10



Note: Figure not drawn to scale.

A right triangle has the dimensions shown, in centimeters (cm). What is the area of this triangle, in cm^2?

Question N. 11



In the figure, line AB is parallel to line CD, and line AD intersects line BC at point E. If $y = 119$ and $z = 113$, what is the value of x?

A) 52

B) 67

C) 128

D) 142

Question N. 12

A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week, the number of prey had increased by 1900% of the number of prey at the start of the week, and the number of predators had increased by 240% of the number of predators at the start of the week. The number of predators at the end of the week was p% less than the number of prey at the end of the week. What is the value of p?

Question N. 13

Which expression is **NOT** a factor of $6{,}480x^4 - 1{,}280$?

A) $9x^2 + 4$

B) $3x^2 - 2$

C) $3x + 2$

D) 80

Question N. 14

$$-4x + 24px = 72$$

In the given equation, p is a constant. The equation has no solution. What is the value of p?

A) 0

B) $\frac{1}{6}$

C) $\frac{3}{4}$

D) 3

Question N. 15

A zoologist observed the nests of wood ducks in an area. Each year, the zoologist recorded the number of eggs in each of the first 35 nests that were observed and created a frequency table for the data set. Which of the following frequency tables represents the data set with the smallest standard deviation?

A)

Number of eggs	Frequency
10	8
11	7
12	5
13	7
14	8

B)

Number of eggs	Frequency
10	7
11	7
12	7
13	7
14	7

C)

Number of eggs	Frequency
10	2
11	8
12	15
13	8
14	2

D)

Number of eggs	Frequency
8	0
9	5
10	35
11	5
12	0

Question N. 16

$$(x+2)(x-12)=0$$

What is the positive solution to the given equation?

Question N. 17

The function f is a quadratic function. In the xy-plane, the graph of $y=f(x)$ has a vertex at $(1,4)$ and passes through the points $(2,28)$ and $(-1,100)$. What is the value of $f(-2)-f(0)$?

A) 76

B) 124

C) 192

D) 220

Question N. 18

$$2x^2+14x+2y^2+20y=\frac{15}{2}$$

The given equation defines a circle in the xy-plane. What is the radius of the circle?

A) $\dfrac{\sqrt{30}}{2}$

B) $\sqrt{15}$

C) $\sqrt{41}$

D) $\dfrac{\sqrt{626}}{2}$

Question N. 19

$$y = 6x^2 - 48x + 99$$
$$y + 4 = 0$$

How many solutions are there to the given system of equations?

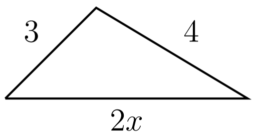
A) There is exactly 1 solution.

B) There are exactly 2 solutions.

C) There are exactly 3 solutions.

D) There are no solutions.

Question N. 20



Note: Figure not drawn to scale.

The triangle in the figure above has side lengths 3, 4, and $2x$. Which of the following is NOT a possible value of x?

A) 1

B) 2.5

C) 3

D) 5

Question N. 21

Line p in the xy-plane contains the points $(6, 0)$ and $(6, 6)$. Which equation defines line p?

A) $x = 0$

B) $x = 6$

C) $y = 0$

D) $y = 6$

Question N. 22

The function f is defined by $f(z) = 3z$. Which of the following is equivalent to $f(z-7)$?

A) $-3z$

B) $3z - 7$

C) $3z - 21$

D) $-3z + 21$

Module 2

Question N. 1

The expression $\dfrac{kx-17}{2x^2-11x+15}$ is equivalent to $\dfrac{p}{x-3}-\dfrac{1}{2x-5}$, where k and p are constants. What is the value of p?

A) -16

B) $\dfrac{14}{5}$

C) 4

D) 7

Question N. 2

The measure of angle F is $\dfrac{\pi}{12}$ radians. If the measure of angle F is $3n$ degrees, where n is a constant, what is the value of n?

A) 3

B) 5

C) 12

D) 15

Question N. 3

The function g is defined by $g(x) = -2x(x+4)(x-k)^2 + r$, where k and r are integer constants. In the xy-plane, the graph of $y = g(x)$ passes through the point $(8, 17)$, and $g(0) = 17$. What is the value of $r + k$?

A) -8

B) 0

C) 17

D) 25

Question N. 4

The function f is defined by $f(x) = 41(0.26)^x$. For any positive integer n, the value of $f(n)$ is $p\%$ less than the value of $f(n-1)$. What is the value of p?

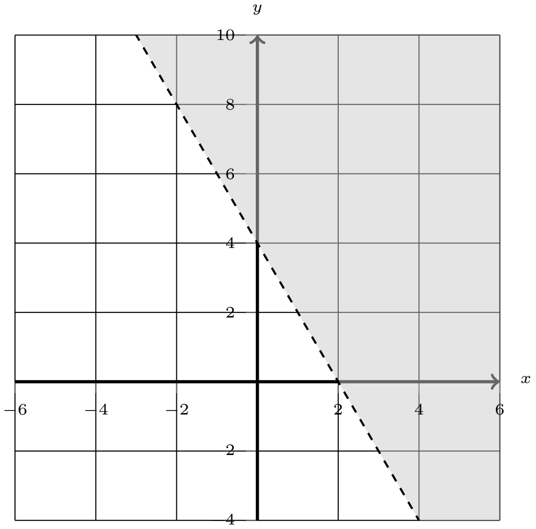
A) 74

B) 59

C) 41

D) 26

Question N. 5



The shaded region shown represents the solutions to which inequality?

A) $y \geq -2x + 4$

B) $y > 4x - 2$

C) $y \leq -2x + 4$

D) $y \leq 4x - 2$

Question N. 6

$$f(x) = 21(1.20)^{\frac{x}{3}}$$

For the given function f, the value of $f(x)$ increases by $p\%$ for every increase of x by 6. What is the value of p?

Question N. 7

A company developed a plan to set the selling price of a product. The company determined that for a selling price of \$90.00, zero products would be sold. For each \$1.50 decrease in the selling price, the number of products sold would increase by one. For a revenue of exactly \$1,336.50, which of the following could be the number of products sold? (revenue = price × number of products sold)

A) 27

B) 30

C) 831

D) 1,350

Question N. 8

$$y = 4x^2 - 24x + 38$$
$$y + 3 = 0$$

How many solutions are there to the given system of equations?

A) There is exactly 1 solution.

B) There are exactly 2 solutions.

C) There are exactly 3 solutions.

D) There are no solutions.

Question N. 9

$$-7x^2 + bx - 63 = 0$$

In the given equation, b is a positive integer. The equation has no real solution. What is the largest possible value of b?

Question N. 10

If $5 + \dfrac{2(13 - x)}{7} = 3(13 - x) + 5$, what is the value of $13 + x$?

Question N. 11

The area of a rectangle is 60 square inches. The length of the longest side of the rectangle is 15 inches. What is the length, in inches, of the shortest side of this rectangle?

A) 4

B) 15

C) 30

D) 45

Question N. 12

For the linear function g, $g(9) = 11$ and $g(x + 1) - g(x) = 2$. Which equation defines g?

A) $g(x) = 9x + 2$

B) $g(x) = 9x + 11$

C) $g(x) = 2x - 18$

D) $g(x) = 2x - 7$

Question N. 13

x	$f(x)$
-9	78
9	-156
27	-390

For the linear function f, the table shows three values of x and their corresponding values of $f(x)$. Function f is defined by $f(x) = rx + s$, where r and s are constants. What is the value of rs?

Question N. 14

$$57x^{18} + bx^9 + 34$$

The given expression, where b is a constant, is equivalent to $(3x^9 + q)(rx^9 + 2)$, where q and r are constants. What is the value of b?

Question N. 15

$$g(x) = x^3 + ax^2 + bx + c$$

The function g is defined by the given equation, where a, b, and c are integer constants. The zeros of the function are -2, -7, and 3. What is the value of a?

A) -42

B) -6

C) 6

D) 42

Question N. 16

A circle in the xy-plane has its center at $(-4, 4)$ and has a radius of 6. An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a, b, and c are constants. What is the value of c?

Question N. 17

A model estimates that the population of a state was 4 million in 1980 and increased by 0.041 million each year until 2007. According to the model, what was the estimated population of this state, in millions, 10 years after 1980?

Question N. 18

In triangle RST, angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 63 units, the length of $\overline{LK}$ is 21 units, and the area of triangle RST is 252 square units, what is the length of $\overline{KT}$, in units?

Question N. 19

A car dealership has only sedans, SUVs, and minivans for sale. On Monday, 20% of the vehicles for sale were sedans and 50% were SUVs. If there were 62 sedans for sale at the dealership on Monday, how many minivans were for sale?

A) 93

B) 310

C) 9

D) 36

Question N. 20

An exponential function f gives the estimated amount of an X-ray beam's initial intensity remaining after passing through a w-centimeter-thick window made of beryllium. The function estimates that after passing through a 1.8-centimeter-thick window, the amount of the X-ray beam's initial intensity remaining is 0.65. Which equation could define f?

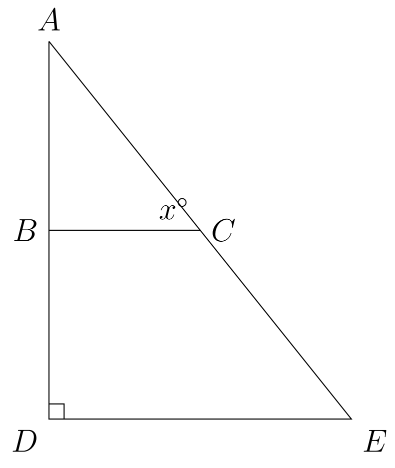
A) $f(w) = 0.65(0.79)^{w-1.8}$

B) $f(w) = 0.65(0.79)^{\frac{w}{1.8}}$

C) $f(w) = 0.65(1.8)^{w}$

D) $f(w) = 1.8(0.65)^{w}$

Question N. 21



Note: Figure not drawn to scale.

In the figure, $\overline{BC}$ is parallel to $\overline{DE}$. If the length of $\overline{DE}$ is 162 and the length of $\overline{AE}$ is 270, what is the value of $\tan x°$?

A) $\frac{270}{162}$

B) $\frac{216}{162}$

C) $\frac{162}{270}$

D) $\frac{162}{216}$

Question N. 22

The function f is defined by $f(x) = |x - 6x|$. What value of a satisfies $f(1) - f(a) = -15$?

A) -16

B) 3

C) 4

D) 75

Answer Sheet

Module 1

1	2	3	4	5	6	7	8	9	10
D	C	64	B	B	A	C	A	C	12

11	12	13	14	15	16	17	18	19	20
C	83	B	B	D	12	C	C	D	D

21	22
B	C

Module 2

1	2	3	4	5	6	7	8	9	10
C	D	D	A	A	44	A	D	41	26

11	12	13	14	15	16	17	18	19	20
A	D	507	36	C	-4	4.41	112/21	A	A

21	22
B	C